

Martin's Life Work

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Chapter 1

Electromagnetism

1.1 Debye-Huckel Theory

Debye-Huckel theory is a description of the electromagnetic interactions between ions in a solution of ionic compounds.

1.1.1 Poisson-Boltzmann

The potential in a field of ions must be solved in terms of electrostatics in a media. This is described by Poisson's equation

$$\nabla^2 \Psi(\mathbf{r}) = -4\pi \frac{\rho(\mathbf{r})}{\varepsilon} \quad (1.1)$$

In a sea of ions, we must carefully define the charge density as it pertains to the system. In a general solution of ions, there will be j species of ions.

$$\nabla^2 \Psi(\mathbf{r}) = -4\pi \frac{\rho_j(\mathbf{r})}{\varepsilon}$$

The charge distribution of a single ion concentration is essentially the density of the ionic positions. It would be tedious and unfeasible to count every ion's position exactly as we are assuming too many things. Rather, one can use a statistical ensemble of ions via a canonical ensemble, which is a good approximation here.

$$\rho_i(\mathbf{r}) = C_i e_0 Z_i \exp[-\beta Z_i e_0 \Psi(r)] \quad (1.2)$$

Here the energy of the system is defined as the potential energy scaled by the charge defined as $U = qV$. The full equation is known as the Poisson-Boltzmann equation, where $\Psi(\mathbf{r})$ is the potential for all space.

$$\nabla^2 \Psi(\mathbf{r}) = -\frac{1}{\varepsilon_0 \varepsilon_r} \sum_{i=1}^{\infty} C_i e_0 Z_i \exp[-\beta Z_i e_0 \Psi(r)] \quad (1.3)$$

There is a nice approximation we can make given that the potential of the solution is small. We linearize the exponential to first order here.

$$\begin{aligned}
\nabla^2 \Psi(\mathbf{r}) &\approx -\frac{1}{\varepsilon_0 \varepsilon_r} \sum_{i=1}^{\infty} C_i e_0 Z_i [1 - \beta Z_i e_0 \Psi(r)] \\
&= -\frac{1}{\varepsilon_0 \varepsilon_r} \sum_{i=1}^{\infty} [C_i e_0 Z_i - \beta C_i Z_i^2 e_0^2 \Psi(r)] \\
&= -\sum_{i=1}^{\infty} \left[\frac{C_i e_0 Z_i}{\varepsilon_0 \varepsilon_r} - \kappa^2 \Psi(r) \right]
\end{aligned}$$

Where κ_i^2 is related to the debye-length, a unit of the electrostatic screening caused by the ionic interactions. The Debye length is defined as $\lambda_D = \kappa^{-1}$.

$$\kappa_i^2 = \frac{\beta C_i Z_i^2 e_0^2}{\varepsilon_0 \varepsilon_r} \quad (1.4)$$

1.1.2 Monovalent Ionic Mixture

Suppose there is a single type of ion in the mixture. The sum collapses into a single term and the constant term is dropped.

$$\nabla^2 \Psi(\mathbf{r}) = \kappa^2 \Psi(\mathbf{r}) \quad (1.5)$$

This equation has an obvious solution which are the real exponentials;

$$\Psi(\mathbf{r}) = A e^{-\kappa r} + B e^{\kappa r} \quad (1.6)$$

Chapter 2

Measure Theory

2.1 Measurable Spaces

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2.1.1 Problems

Problem 2.1. *Partition generated by a σ -algebras.*

1. Let $\mathcal{C} = \{A, B, C\}$ be a partition of E . List all the elements of $\sigma\mathcal{C}$
2. let $\mathcal{C}$ be a countable partition of E . show that every element of $\sigma\mathcal{C}$ is a countable union of elements taken from $\mathcal{C}$.
3. Let $E = \mathbb{R}$. Let $\mathcal{C}$ be the collection of all singleton subsets of $\mathbb{R}$, that is, sets that only contain a single point. Show that each element of $\sigma\mathcal{C}$ is either countable, or the complement of a countable set.

Solution.

1. The elements of $\sigma\mathcal{C}$ are defined as the intersection of all sigma-algebras that contain $\mathcal{C}$. Since $\mathcal{C}$ contains 3 subsets of E , as it is a partition, which implies that A, B, C are disjoint and have a union of E . Thus, we can define $\sigma\mathcal{C}$ as the following set:

$$\sigma = \{\emptyset, E, \{A\}, \{B\}, \{C\}, \{A \cap B\}, \{A \cap C\}, \{B \cap C\}\} \quad (2.1)$$

This set contains E , all partitions of closures by E generated from $\mathcal{C}$. Thus, this is the smallest sigma-algebra that contains E generated by $\mathcal{C}$.

2. Since $\mathcal{C}$ is a partition, $\bigcup \mathcal{C} = E$. We need to show that $\sigma\mathcal{C}$ contains only elements from $\mathcal{C}$. We can take ε as the set of all countable unions taken from E . We can show that this is a sigma-algebra by taking its definitions. Since ε contains E the subset of all the partitions, it must also contain the union of none of the partitions, or $\emptyset$. Secondly, by taking all of the unions, we start with the disjoint partitions and then create their unions until we reach E . This will be closed under complement as by taking countable unions, we will have the sets that are not part of the union as another subset of a different union. Thus, we can argue that this is the same as $\sigma\mathcal{C}$ by the logic of construction. Thus, since each element of $\sigma\mathcal{C}$ is isomorphic to the construction that we obtained from the union of subsets, we have shown that all elements are just a set of unions in $\mathcal{C}$.

3. Define the set $\mathcal{A} = \{A \subset \mathbb{R} : A \text{ is countable or } A^c \text{ is countable}\}$. We want to show that this is a sigma-algebra. Since $\emptyset$ is countable, it is contained within $\mathcal{A}$. $\emptyset \in \mathcal{A}$. The complement of this is the set of reals; $\mathbb{R} = \emptyset^c \in \mathcal{A}$. By definition, if A is countable, then A^c is co-countable. Similarly, if A is co-countable, this implies that A^c is countable. Thus, both sets are contained in $\mathcal{A}$.

Thus, consider a sequence $(A_n)_{n \in \mathbb{N}} \subset \mathcal{A}$. the union of this sequence is $\bigcup_n A_n$. This is split into cases. First, if all A_n are countable, so it the union, which is included in A_n . If all A_n are co-countable, then consider the compliment of the union $(\bigcup_n A_n)^c = \bigcap_n A_n^c$ which is an intersection of countable sets, which is countable. If there is a mix between countable and co-countable sets, they are co-countable, which are contained in A by the previous definition. Thus, $\bigcup_n A_n \in \mathcal{A}$. Thus, $\mathcal{A}$ is closed under union and complement. Thus, $\mathcal{A}$ is a sigma-algebra. Thus, $\sigma\mathcal{C} \subset \mathcal{A}$

We want to show that any countable set is a union of the singletons. Since A contains countable and co-countable sets, A is a union of singletons, which are by definition contained in $\mathcal{A}$.

$$A = \bigcup_{x \in A} \{x\} \quad (2.2)$$

For co-countable sets in A , we note that the complements are countable, which are contained in the above definition. Since $\mathcal{A}$ consists of countable sets, these are the countable unions of singletons contained in $\sigma\mathcal{C}$. Since $\mathcal{A}$ also consists of co-countable sets, these are the complements of intersections of countable sets in $\sigma\mathcal{C}$. Thus, $\mathcal{A} \subset \sigma\mathcal{C}$.

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Problem 2.2. Let (E, ε) be a measurable space. Fix $D \subset E$ and let

$$\mathcal{D} = \varepsilon \cap D = A \cap D : A \in \varepsilon$$

Show that $\mathcal{D}$ is a sigma-algebra on D . It is called the trace of (E, ε) on D .

Solution. To show that $\mathcal{D}$ is a sigma-algebra, we need to show closure under union, complement, intersection, and the existence of $D, \emptyset \in \mathcal{D}$.

1. To show that $D, \emptyset \in \mathcal{D}$ i will write out the logical steps. Since $D \subset E$, $D \in \varepsilon$. Since $A = D$, $D \cap D = D$. By definition of $\mathcal{D}$, $D \in \mathcal{D}$. Similarly, $\emptyset \subset E$, $\emptyset \in \varepsilon$. Since $A = \emptyset$, $\emptyset \cap D = \emptyset$. By definition of $\mathcal{D}$, $\emptyset \in \mathcal{D}$.
2. The closure of complement can be shown by the set property of D^c . Since $E^c = \emptyset$, $D^c = E \setminus D$.

$$\begin{aligned} (A_n \cap D)^c &= A^c \cup D^c \\ &= A^c \cup (E \setminus D) \\ &= E \setminus (D \cap A) \\ &\in \mathcal{D} \end{aligned}$$

Thus $(A_n \cap D)^c \in \mathcal{D}$.

3. For union and intersection we will use a similar set operation;

$$\begin{aligned} \left(\bigcup_{i=1}^{\infty} A_i \right) \cap D &= \lim (A_1 \cup A_2 \cup \dots \cup A_n) \cap D \\ &= \lim (A_1 \cap D) \cup (A_2 \cap D) \cup \dots \cup (A_n \cap D) \\ &= \bigcup_{i=1}^{\infty} (A_i \cap D) \\ &\in \mathcal{D} \end{aligned}$$

$$\begin{aligned}
\left(\bigcap_{i=1}^{\infty} A_i\right) \cap D &= (A_1 \cap A_2 \cap \cdots \cap A_n) \cap D \\
&= \lim (A_1 \cap D) \cap (A_2 \cap D) \cap \cdots \cap (A_n \cap D) \\
&= \bigcap_{i=1}^n (A_i \cap D) \\
&\rightarrow \left(\bigcap_{i=1}^{\infty} (A_i \cap D)\right)^c \\
&= \bigcup_{i=1}^{\infty} (A_i \cap D)^c \\
&= \bigcup_{i=1}^{\infty} (A_i^c \cup D^c)
\end{aligned}$$

By the closure of the complement: $\in \mathcal{D}$

Thus we have shown that $\mathcal{D}$ is a sigma-algebra on D .

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Problem 2.3. Let (E, ε) be a measurable space, let $\{\Delta\}$ be a single-point not in E . Let $\bar{E} = E \cup \{\Delta\}$. Show that

$$\hat{\varepsilon} = \varepsilon \cup \{A \cup \{\Delta\} : A \in \varepsilon\}$$

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Problem 2.4. Let $(E, \varepsilon), (F, \mathcal{F})$ be measurable spaces. Show that the product $\varepsilon \otimes \mathcal{F}$ is also the sigma-algebra generated by $\hat{\varepsilon} \cup \mathcal{F}$ where

$$\hat{\varepsilon} = \{A \times F : A \in \varepsilon\} \quad \mathcal{F} = \{E \times A : A \in \mathcal{F}\} \quad (2.3)$$

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2.2 Measureable Functions

Measureable functions are defined as maps between Measureable spaces. Let $(E, \mathcal{E})$ and $(F, \mathcal{F})$ be measurable spaces. A measurable function is defined as

$$f : E \rightarrow F, \quad f^{-1}B \in \mathcal{E} \quad \forall B \in \mathcal{F}$$

2.2.1 Problems

Problem 2.5. Let E be a set and $(F, \mathcal{F})$ a measurable space. For $f : E \rightarrow F$, define

$$f^{-1}\mathcal{F} = \{f^{-1}B : B \in \mathcal{F}\}$$

Where $f^{-1}B = \{x \in E : f(x) \in B\}$. Show that $f^{-1}\mathcal{F}$ is a sigma-algebra on E .

Solution. Since $f^{-1}\mathcal{F} = \{f^{-1}B : B \in \mathcal{F}\}$ we need to show the properties. Since $f^{-1}F = E$, $E \in f^{-1}\mathcal{F}$. Similarly $f^{-1}\emptyset = \emptyset$, $\emptyset \in f^{-1}\mathcal{F}$. To show complement, consider $A \in \mathcal{F}$. Then $A^c \in \mathcal{F}$. Thus, we require by definition that $f^{-1}A^c \in E$. Thus the set is closed under complement for all $A \in \mathcal{F}$. For Union and intersection we use the definition;

$$\left(\bigcup_{i=1}^{\infty} A_i\right) \in \mathcal{F}$$

$$f^{-1}\left(\bigcup_{i=1}^{\infty} A_i\right) \in E$$

$$\left(\bigcap_{i=1}^{\infty} A_i\right)^c = \bigcup_{i=1}^{\infty} A_i^c$$

$$\in \mathcal{F}$$

$$f^{-1}\bigcap_{i=1}^{\infty} A_i \in E$$

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2.3 Measures

Let $(E, \mathcal{E})$ be a measurable space. A measure on $(E, \mathcal{E})$ is defined as a map $\mu : \mathcal{E} \rightarrow \bar{\mathbb{R}}_+$ where $\bar{\mathbb{R}}_+ = [0, +\infty]$. The measure has properties;

$$\mu(\emptyset) = 0$$

$$\mu\left(\bigcup_n A_n\right) = \sum_n \mu(A_n) \text{ for every disjointed sequence } (A_n) \in \mathcal{E}$$

The definition of the measure contains several properties;

1. If $A \cap B = \emptyset$ Then $\mu(A \cup B) = \mu(A) + \mu(B)$
2. If $A \subset B$ Then $\mu(A) \leq \mu(B)$
3. If $A_n \nearrow A$ Then $\mu(A_n) \nearrow \mu(A)$
4. $\mu(\bigcup_n A_n) \leq \sum_n \mu(A_n)$

Some examples of measures are listed below for the measurable space $(E, \mathcal{E})$;

1. Dirac Measure:

$$\delta_x(A) = \begin{cases} 1 & x \in A \\ 0 & x \notin A \end{cases}$$

2. Counting Measure: Let D be a fixed subset of E . The counting measure counts the number of points $A \cap D$. We interpret this as all elements outside D have zero measure and all elements of D have equal measure.

$$\#_D(A) = \sum_{x \in D} \delta_x(A), \quad A \in \mathcal{E}$$

3. Discrete Measure: Let D be a countable subset of E , and $m(x)$ a positive number.

$$\mu(A) = \sum_{x \in D} m(x) \delta_x(A), \quad A \in \mathcal{E}$$

2.3.1 Problems

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2.4 Integration

Let $(E, \mathcal{E}, \mu)$ be a measure space. Here we define $\mathcal{E}$ as the sigma algebra of all $\mathcal{E}$ -measurable functions, which are all measurable functions mapping to the Borel sigma-algebra.

We define the 'integral of f with respect to μ ' as:

$$\mu f = \mu(f) = \int_E \mu(dx) f(x) = \int_E f d\mu \quad (2.4)$$

Definition 2.1. Let f be simple and positive, such that it has the form $f(x) = \sum_{i=1}^n a_i 1_{A_i}(x)$. Then we define

$$\mu f = \sum_{i=1}^n a_i \mu(A_i)$$

This definition can be used on different measures to see how the integral is defined.

1. Dirac Measure:

$$\mu f = \delta_x(A) f(x) = f(A)$$

2. Discrete Measure:

$$\mu f = \sum_{x \in D} m(x) f(x)$$

3. Lebesgue Integrals:

$$\mu f = \text{Leb}_E f = \int_E \text{Leb}_E(dx) f(x) = \int_E dx f(x)$$

Note that if the lebesgue exists, so does the riemann integral, which is reminded by the notation in the last equality. Note that the converse is false.

The main idea of the integral with respect to a measure is packaged in how the measure acts on infinitesimal pieces dx . Since the measure gives different weights to different parts of the set. We can use a result that approximates a measurable function f as a convergent sequence of simple functions. That is, we can approximate f using simple functions that partition the domain in a way that converges to the value of the integral.

Chapter 3

Probability

Problem 3.1. *Given N people in a room, what is the probability that 2 or more people share a birthday?*

Solution. This problem is an exercise in counting. First, let's go case-by-case and write out the general solution. There are 365 days in total. If there is only 1 person in the room, only 1 day is taken. Once we place the second person in the room, we can calculate the probability of the two individuals not-sharing the same birthday;

$$\mathbb{P}[\text{share birthday}] = 1 - \mathbb{P}[\text{not share birthday}] \quad (3.1)$$

I.e., the second person has only 364 days to choose from to not share. Thus, their probability of not sharing a birthday are 364/365, i.e. the event space over the sample space. Likewise, the following person has now only 363 days to choose from and so on. This relation can be multiplied together such that the overall probability of not sharing is that nobody in the room is sharing. This can be built using a product;

$$\mathbb{P}[\text{not share birthday}] = \prod_{x=1}^N \frac{365 - x}{365} \quad (3.2)$$

This gives us the probability that at least 2 people are sharing;

$$\mathbb{P}[\text{at least 2 share birthday}] = 1 - \prod_{x=1}^N \frac{365 - x}{365} \quad (3.3)$$

Plotting this relation as a function of N , we can see the shape of this relation in Fig. 3.1.

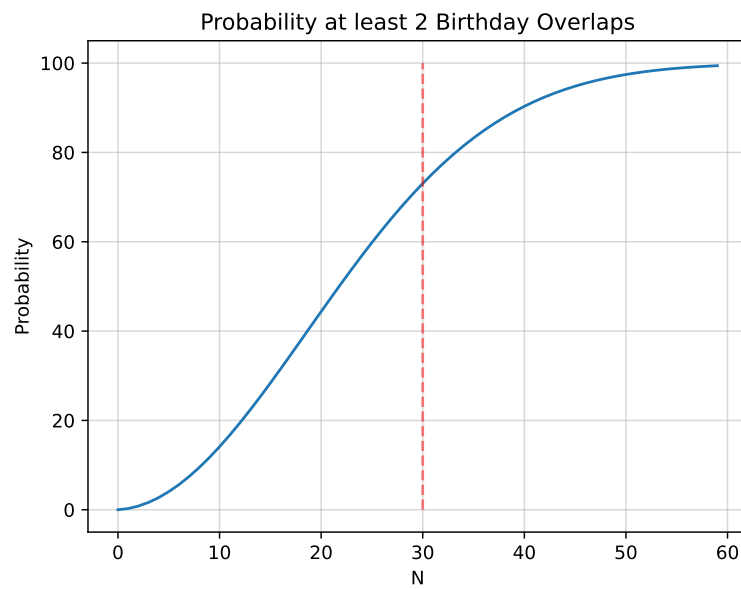


Figure 3.1: Probability of at least 2 people in a room sharing a birthday as a function of N people in the room. at $N = 30$, $\mathbb{P} \approx 73\%$

Chapter 4

Digital Image Theory and Application

I have been working on images at my lab for quite some time now. This is a summary and documentation of the things i have learned while on the job.

4.1 Images as a Hilbert Space

At its core, images are taken by a camera array, which contains *pixels*. Each pixel is one point of the image, or a binning of millions of photons hitting the image, creating a smooth description of some object being photographed. Definitions of the image can be done in 2 ways; theoretical and practical.

Definition 4.1. *Let $f(x, y, z)$ be a field in $\mathbb{R}^3$. Let $\hat{P}$ be the projection operator such that $\hat{P} : \mathbb{R}^3 \rightarrow L_2(\mathbb{R}^2)$. Then the digital image is defined as $g(x, y) = \hat{P}f(x, y, z)$.*

the results of this defintion imply that the dimension z is integrated out. For instance, images of physical objects are the integration of the density function of space, $\rho(x, y, z)$, or when using a light emission system, the optical field density $I(x, y, z)$.

This defintion of the image lives in the vector-space of $\mathbb{R}^2$, which allows for a set of known transformations. Affine transformations are defined by a transformation matrix $\mathbf{T}$ and a shift $\vec{v}$.

$$f_T(\vec{x}) = f(\mathbf{T}^{-1}(\vec{x}) + \vec{b}) \quad (4.1)$$

For group transformations, we can define the action $(g \circ f)(\vec{x}) = f(g^{-1}\vec{x})$. This definition implies linearity for sums of scalar transformations of images;

$$g(af_1 + bf_2) = a(g \circ f_1) + b(g \circ f_2) \quad (4.2)$$

Furthermore, we can define the group as being part of $SO(2)$. This group includes the rotation matrices, which gives the norm-preservation property;

$$|g \circ f| = |f|_{L_2} \quad (4.3)$$

Naturally, the functions we produce want to have some sense of similarity between them. Ie, how similar are two images? To do this, we can re-use the idea of a generalized angle. Two images can be the same in terms of angle, but have different magnitudes. This can be interpreted as a measure in distance. From QM, we know that a square-integratable set of functions with angle and distance are defined in the Hilbert space.

Definition 4.2. *Let $\mathcal{H}$ denote the Hilbert space; Elements of $\mathcal{H}$ must satisfy the following criteria. Let $a, b \in \mathcal{H}$:*

1. Elements have an inner product: $\langle a, b \rangle$
2. Norm Property: $\langle a, a \rangle > 0$
3. Zero Property: $\langle a, a \rangle = 0$ if $x = 0$
4. Inner product commutation: $\langle a, b \rangle = \langle b, a \rangle^*$
5. Linearity for constants: $\langle a, e_1 b_1 + e_2 b_2 \rangle = e_1^* \langle a, b_1 \rangle + e_2^* \langle a, b_2 \rangle$
6. Angle: $A(a, b) = \langle a, b \rangle$
7. Distance: $d(a, b) = \sqrt{\langle a - b, a - b \rangle}$

The inner product of two functions is evaluated as their integral over the domain, hence;

$$\langle f(x, y), g(x, y) \rangle = \iint f g^* dx dy \quad (4.4)$$

From QM, we know about the existence of unitary operators. Below is a definition that we can use to define rotation operations.

Definition 4.3. Let U be a unitary operator acting on f . $U_\theta : L_2(\mathbb{R}^2) \rightarrow L_2(\mathbb{R}^2)$ such that the norm is preserved $|U_\theta f| = |f|$.

4.1.1 Rotational Symmetry

Let U_θ be a unitary rotation operator in $SO(2)$.

We can define exact symmetry as an identical copy due to rotations; $U_\theta f = f$. We know that our images will contain some noise; $U_\theta(f + \eta) = f + U_\theta \eta$. These disturbances will not be symmetrical.

Definition 4.4. Let the rotational symmetry be defined as how similar the image is to its rotated counterparts normalized by the norm of the image.

$$S(f) = \sup_\theta \frac{\langle f, U_\theta f \rangle}{|f|^2} = 1 - \inf_\theta \frac{|f - U_\theta f|}{|f|} \quad (4.5)$$

We can also define the maximum symmetry being $S(f) = 1$, whereas minimum symmetry, or a total orthogonal set, $S(f) = 0$.

This function is bounded on $[0, 1]$. The supremum clause defines the largest possible deviations from a perfect symmetry.

This definition uses the auto-correlation of the image with its rotational counter-part, however, integrates over the radial-ness of the image. For parts of the image that are only radially symmetric for a set of angles, we can define the measure over a radial average;

For instance, say that the image is symmetric over a wide range of angles, but contains portions where the symmetry is broken by a mis-alignment or blip, these would be integrated over in the symmetry measure.

Definition 4.5. Let the radial average be defined as the integral over angles θ_0 .

$$f_0(r, \Delta\theta, \theta_0) = \frac{1}{2\pi} \int_{\theta_0}^{\theta_0 + \Delta\theta} f(r, \theta) d\theta \quad (4.6)$$

Then, define the symmetry metric as the ratio between the norm of the image, and the norm of the radial averaged-portion;

$$S(r, \theta_0, \Delta\theta) = \frac{|f_0|^2}{|f|^2} \quad (4.7)$$

Chapter 5

Computer Science and Algortihms

This is mainly about algorithmns as i dont know shit about computers

5.1 Dynamic Programming

5.1.1 2D Chain-Link Algorithm

Algorithm 1 Dynamic Programming for Best 2D Chain of Points

Require: $\text{min_idx} = \{\text{min_idx}[t]\}_{t=0}^{F-1}$, list of candidate positions per frame

1: $\text{min_vals} = \{\text{min_vals}[t]\}_{t=0}^{F-1}$, list of candidate values per frame

2: $w = (w_x, w_y)$, weighting for distance computation (default (1, 1))

3: max_step , optional maximum allowed step distance

Ensure: $\text{idx_chain}, \text{x_chain}, \text{y_chain}$, the indices, positions, and values of the optimal path

4: Initialize $F \leftarrow |\text{min_idx}|$

5: Initialize $\text{cost}[i] \leftarrow 0$ for all i in frame 0

6: Initialize $\text{prev} \leftarrow []$

▷ to store predecessor indices

7: **for** $t = 1$ to $F - 1$ **do**

8: $A_x \leftarrow \text{min_idx}[t - 1]$, $B_x \leftarrow \text{min_idx}[t]$

9: $A_y \leftarrow \text{min_vals}[t - 1]$, $B_y \leftarrow \text{min_vals}[t]$

10: Compute pairwise differences:

$$\Delta_x[i, j] = |A_x[i] - B_x[j]|, \quad \Delta_y[i, j] = |A_y[i] - B_y[j]|$$

11: Compute weighted Euclidean distances:

$$D[i, j] = \sqrt{(w_x \Delta_x[i, j])^2 + (w_y \Delta_y[i, j])^2}$$

12: **if** max_step is not None **then**

13: Set $D[i, j] \leftarrow \infty$ if $D[i, j] > \text{max_step}$

14: **if** no finite $D[i, j]$ exists **then**

15: **return** empty path

16: **end if**

17: **end if**

18: Compute cumulative path cost matrix:

$$M[i, j] = \text{cost}[i] + D[i, j]$$

19: For each node j in frame t , find predecessor $p[j]$ minimizing $M[:, j]$:

$$p[j] = \arg \min_i M[i, j], \quad \text{new_cost}[j] = M[p[j], j]$$

20: Update $\text{cost} \leftarrow \text{new_cost}$

21: Append p to prev

22: **end for**

23: Initialize $j \leftarrow \arg \min_i \text{cost}[i]$ ▷ index of minimal total cost in last frame

24: Initialize $\text{idx_chain} \leftarrow [0]^F$, set $\text{idx_chain}[F - 1] \leftarrow j$

25: **for** $t = F - 2$ down to 0 **do**

26: $\text{idx_chain}[t] \leftarrow \text{prev}[t][\text{idx_chain}[t + 1]]$

27: **end for**

28: Compute chains:

$$\text{x_chain}[t] = \text{min_idx}[t][\text{idx_chain}[t]], \quad \text{y_chain}[t] = \text{min_vals}[t][\text{idx_chain}[t]]$$

29: **return** $\text{idx_chain}, \text{x_chain}, \text{y_chain}$
